

Lagrangian Mechanics But It Makes Sense

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Dedicated to my favorite physics clubber William "Willy" Wang.

Preface

Lagrangian Mechanics... a sort of holy grail in undergraduate physics, or so I've heard. Lagrange's equation itself is not hard to learn to use, as long as you understand partial derivatives. However, the origin of the topic is usually left somewhat ambiguous. Most sources will derive the Euler-Lagrange equation, with a lot of arbitrary definitions and confusing notation. Then there is the choice of the Lagrangian - why is it $\mathcal{L} = T - U$? In short, without a good teacher, it is extremely hard to learn the theory behind Lagrangian Mechanics.

Well I obviously somehow managed, since I'm here writing this handout. After reading too many articles and Classical Mechanics textbook chapters, I think I have a decent grasp on Lagrangian Mechanics. (Hopefully saying that none of this handout is rigorous will cover for any mistakes I make.) And with a good grasp on calculus, some free time to learn a bit of Multivariable Differential Calculus (try MIT OCW 18.02 Lectures, or read the Appendix), and some patience, any high school senior should also be able to understand Lagrangian Mechanics.

Good luck.

1 Motivations for the Calculus of Variations

The calculus of variations is the mathematical foundation that Lagrangian mechanics is built on, so I want to motivate the need to develop Lagrangian mechanics and also the types of problems calculus of variations can solve.

Newton's 2nd Law is the foundation of Newtonian mechanics and it works great in Cartesian coordinates. However, it tends to fall apart in other coordinate systems due to how messy it becomes. Lagrangian mechanics is the alternative which will work with what we call generalized coordinates.

To see how calculus of variations will help us with this goal, let us first see what types of problems it can solve.

1.1 The Catenary

A catenary is a curve that describes how a flexible chain would hang between two poles.



The principle behind the catenary is that it must minimize the total potential energy of the chain, that is, we want to minimize:

$$\int gy dm = \int_C \mu gy ds = \int_{x_1}^{x_2} \mu gy(x) \sqrt{1 + y'(x)^2} dx.$$

Note the fact that y is a function of x and we are finding a function such that the value of this integral is minimized, rather than the typical calculus problem of finding a value that minimizes an expression.

1.2 Fermat's Principle

Another important example is the path that light will take between two points. In the same medium, light will obviously travel in a straight line. However, in media of varying refractive indices, Fermat found that light will take the path that requires the shortest time of travel. Given that the speed light travels in a medium with refractive index n is $v = \frac{c}{n}$, we seek to minimize is:

$$\int_{t_1}^{t_2} dt = \int_C \frac{ds}{v} = \frac{1}{c} \int_C n ds = \frac{1}{c} \int_{x_1}^{x_2} n \sqrt{1 + y'(x)^2} dx$$

Again we are finding a function that minimizes the integral.

1.3 Lagrangian

As we have seen in the previous sections, calculus of variations is designed to tackle problems that find the stationary points of an integral with respect to a function (which happened to be minimization with our examples). It turns out finding the path a system will take in mechanics is also a problem of finding stationary points. We will first use calculus of variations to see how to go about this process of finding stationary points, and then derive Hamilton's principle to find the expression that when stationary, gives us the equations of motion.

2 The Functional and the Variation

2.1 The Functional

Since we are minimizing expressions with respect to a function rather than a variable, we need to introduce the functional. The functional $I[f]$ is a type of function that maps a function to a real number. For example, with Fermat's principle, the functional would map a function representing the path light takes to the amount of time it takes for light to travel that path. Then, making the functional stationary would give us the path that minimizes the time of travel.

2.2 The Variation

Thus, we need to develop calculus tools for the functional. Similar to the derivative of a function, there is the derivative of a functional. However, it is a bit complicated to understand, and not strictly necessary for our purposes, so we will instead turn to the differential.

The rigorous definition of the derivative involves a limit on the local slope of a function, but there is another nonrigorous method involving differentials that gives more intuition on infinitesimal change. We will explore just enough to give insight into how we should define a "differential" for a functional, but I highly recommend Silvanus Thompson's *Calculus Made Easy* for further reading.

Suppose that we have a function $f(x) = x^2$. If we change the x value by a bit, let's say $x \rightarrow x + dx$, then the value of the function also changes, from $f \rightarrow f + df$. We can use this to find an expression for df :

$$\begin{aligned}f + df &= f(x + dx) \\df &= f(x + dx) - f(x) \\df &= (x + dx)^2 - x^2 \\df &= 2xdx + (dx)^2 \\df &= 2xdx.\end{aligned}$$

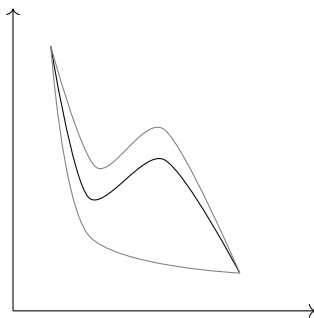
Note that we take a first order approximation and drop the $(dx)^2$ term, as it's negligible compared to dx . In general, the differential can be seen as a infinitesimal nudge to the input. Notice how the derivative appears in the equation. While we were able to directly solve for the differential in this case since we had an explicit function, it is not always the case, and more generally we have,

$$df = \frac{df}{dx}dx.$$

We are now ready to construct a similar definition for the functional. To do this we will introduce the variation, represented by δ . Suppose for a functional $I[f]$, we vary the input function by $f \rightarrow f + \delta f$. Then naturally the output should also vary by $I \rightarrow I + \delta I$. In other words, we can define the variation of a functional by:

$$\delta I = I[f + \delta f] - I[f].$$

Let us first think about what it means to vary the input function of a functional. With a normal function, varying the input means either increasing or decreasing the value of the input, but with a function, there are infinitely many ways to vary it. Below are a few examples of what $f + \delta f$ could look like relative to f :



We can stretch our function in infinitely many different ways, but we still have to be able to control how much we stretch it. This motivates the following definition for the variation of a function:

$$\delta f(x) = \epsilon \eta(x),$$

where $\eta(x)$ is an arbitrary function that determines how it will change the shape of the original function, and ϵ is a parameter that determines how much it will change.

Varying ϵ is much easier than directly varying f because of two main reasons. First, regardless of the η we choose, ϵ tells us how close we are to the original function. The smaller the magnitude of ϵ , the closer to the original function. At $\epsilon = 0$, we get the original function. Second, we have already developed the tools for varying a function with respect to a variable: namely, derivatives.

Now, let us return to finding the variation of a functional using ϵ . If we change ϵ by a little bit, that changes f , which then changes $I[f]$. Since our goal is to find how much $I[f]$ changed, we can write this in terms of rates of change, that is:

$$\delta I = \left. \frac{dI[f + \epsilon \eta]}{d\epsilon} \right|_{\epsilon=0} \epsilon.$$

Note that we evaluate the derivative at $\epsilon = 0$ since we're looking for the variation of I when we vary the original function. Another way to arrive at this expression is a first-order Taylor expansion of $I[f + \epsilon \eta] - I[f]$.

Additional note: It may be tempting to think of the derivative in the equation above as the functional derivative, and although it is closely related, it is not quite the same.

2.3 Properties of the Variation

In some ways, the variation behaves similarly to the differential, but other times not. Here we will list out the two main properties we will use for calculations. No attempt at a proof will be made.

1. $f\delta x = \delta F$ where F is the antiderivative of f . This property is a case where the variation and differential work similarly. For example, $x\delta x = \delta(\frac{1}{2}x^2)$.
2. $(\delta f)' = \delta(f')$. Self-explanatory.

2.4 Stationary Points

With derivatives, stationary points are quite easy to understand. If you think of derivatives as rates of change, stationary points must occur at points where the instantaneous rate of change is 0. If you think of derivatives as the slope of a tangent line, it must be where the tangent line is flat. However, what does this mean in terms of differentials?

At a stationary point, if you change the value of the input variable, the value of the function will likely change, but if you change the value of the input by an infinitesimal amount, then the value of the function will stay the same. This is what is meant when the *instantaneous* rate of change is 0, or the tangent line is flat. Locally at the stationary point, the value of the function is constant.

The same analogy can be applied to functionals. Locally at a stationary point of a functional, if you change the input function infinitesimally, the value of the functional does not change, or in other words, the variation of the functional is 0 at any stationary point. Thus, to find the stationary points of any functional, the following condition has to be satisfied:

$$\delta I = 0 \implies \left. \frac{dI[f + \epsilon\eta]}{d\epsilon} \right|_{\epsilon=0} = 0.$$

3 The Euler-Lagrange Equation

Suppose we take the catenary example from before and decide to minimize the integral. Let us denote the integrand by

$$f[x, y(x), y'(x)] = \mu g y(x) \sqrt{1 + y'(x)^2}.$$

The choice of parameters of the functional seems obvious for this example as it only depend on these three, but it is completely possible to derive a different Euler-Lagrange equation by taking higher derivatives as parameters. As we will see later with Hamilton's Principle, this is the most useful form.

Thus, we have to minimize the integral:

$$S = \int_{x_1}^{x_2} f[x, y(x), y'(x)] dx,$$

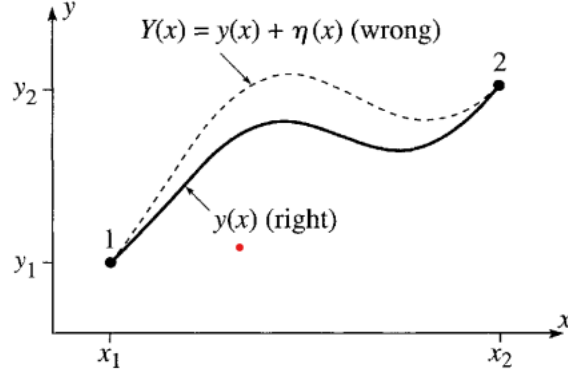
which means we must have $\delta S = 0$, or

$$\left. \frac{dS[y + \epsilon\eta]}{d\epsilon} \right|_{\epsilon=0} = 0.$$

We look at the variation of S by looking what happens when we vary $y(x)$, which also affects $y'(x)$. For convenience, we will define the following:

$$Y(x) = y(x) + \epsilon\eta(x)$$

$$Y'(x) = y'(x) + \epsilon\eta'(x).$$



We also know that the endpoints of our function should remain the same when we vary it. For example, with the catenary, it makes sense that the endpoints of the chain is fixed and only the shape unknown. Thus, we have

$$y(x_1) = y_1, y(x_2) = y_2$$

$$\eta(x_1) = \eta(x_2) = 0.$$

Now we can return to evaluating the derivative. We write S as an integral of f and exchange the order of the derivative and integral, giving:

$$\left. \frac{dS[y + \epsilon\eta]}{d\epsilon} \right|_{\epsilon=0} = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \int_{x_1}^{x_2} f[x, Y(x), Y'(x)] dx = \int_{x_1}^{x_2} \left. \frac{\partial f}{\partial \epsilon} \right|_{\epsilon=0} dx.$$

Applying chain rule, we get

$$\frac{\partial f}{\partial \epsilon} = \frac{\partial f}{\partial Y} \frac{\partial Y}{\partial \epsilon} + \frac{\partial f}{\partial Y'} \frac{\partial Y'}{\partial \epsilon} = \eta \frac{\partial f}{\partial Y} + \eta' \frac{\partial f}{\partial Y'}.$$

Evaluating the derivative at $\epsilon = 0$, Y and Y' become y and y' , respectively, and we get:

$$\left. \frac{\partial f}{\partial \epsilon} \right|_{\epsilon=0} = \eta \frac{\partial f}{\partial y} + \eta' \frac{\partial f}{\partial y'}.$$

We now plug this expression back into the integral for further manipulation. We will integrate the second term first by using integration by parts. We will use a variation of it that is not as quite commonly seen, that is:

$$\int u'v dx = uv - \int uv' dx.$$

Applying this, we have:

$$\int_{x_1}^{x_2} \eta' \frac{\partial f}{\partial y'} dx = \eta \frac{\partial f}{\partial y'} \Big|_{x_1}^{x_2} - \int_{x_1}^{x_2} \eta \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) dx = - \int_{x_1}^{x_2} \eta \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) dx.$$

The first term on the RHS cancels out because as we mentioned early, the endpoints of the curve must remain the same so η evaluated there must be 0.

Substituting back into the integral, we get the equation:

$$\frac{dS}{d\epsilon} \Big|_{\epsilon=0} = \int_{x_1}^{x_2} \eta \left[\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right] dx = 0.$$

Since $\eta(x)$ is an arbitrary function, this equation must be satisfied for all η , and it follows that

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0.$$

This is the Euler-Lagrange equation.

4 Applications of the Euler-Lagrange

As an example, let's solve the catenary problem and find the equation of a hanging chain. Recall from earlier that we are trying to minimize

$$\int gy dm = \int_C \mu gy ds.$$

Earlier we wrote y as a function of x , but here we will actually write x as a function of y to simplify calculations. Thus, the integral is,

$$\int_{y_1}^{y_2} \mu gy \sqrt{1 + x'(y)^2} dy = \int_{y_1}^{y_2} f[y, x(y), x'(y)] dy.$$

The Euler-Lagrange equation tells us

$$\frac{\partial f}{\partial x} - \frac{d}{dy} \left(\frac{\partial f}{\partial x'} \right) = 0.$$

Since f does not depend on x , we have $\partial f / \partial x = 0$. Thus, we have:

$$\begin{aligned}
\frac{d}{dy} \left(\frac{\partial f}{\partial x'} \right) &= 0 \\
\frac{\partial f}{\partial x'} &= \text{const.} \\
\frac{\mu g y x'}{\sqrt{1 + (x')^2}} &= \text{const.} \\
\frac{y x'}{\sqrt{1 + (x')^2}} &= C.
\end{aligned}$$

Here we can solve for x' and then integrate to find $x(y)$.

$$\begin{aligned}
x' &= \sqrt{\frac{C^2}{y^2 - C^2}} \\
x &= \int \sqrt{\frac{1}{(y/C)^2 - 1}} dy \\
&= \cosh^{-1} \left(\frac{y}{C} \right) + C_1 \\
y &= C \cosh(x) + C_1
\end{aligned}$$

Of course the particular solution depends on initial conditions, but this tells us that a chain will always hang in the shape of a cosh function.

5 Hamilton's Principle

Thus far, we have been developing the mathematical foundations we need for Lagrangian Mechanics, but now we can do the physics. We will relate the Euler-Lagrange equation we already derived to mechanics by showing that a physical law is equivalent to a problem of finding stationary points.

The idea is that since we know we want the Lagrangian formulation of mechanics to be equivalent to the Newtonian formulation, the Euler-Lagrange equation should essentially evaluate to Newton's 2nd Law, and we work backwards from there, reversing the derivation of the Euler-Lagrange equation.

We start by considering a system that moves along a single coordinate with respect of time, let's say $x(t)$. If we assume our system is conservative, we have:

$$F = m\ddot{x} = -\frac{dU(x)}{dx}.$$

We can then multiply both sides of the equation by a small variation in x of δx :

$$m\ddot{x}\delta x = -\frac{dU}{dx}\delta x.$$

Then, moving everything to one side, and integrating with respect to time, we have:

$$\int_{t_1}^{t_2} \left(-m\ddot{x}\delta x - \frac{dU}{dx}\delta x \right) dt = 0.$$

Integrating the first term by integration by parts, we get:

$$\int_{t_1}^{t_2} -m\ddot{x}\delta x dt = -m\dot{x}\delta x \Big|_{t_1}^{t_2} + \int_{t_1}^{t_2} m\dot{x}\delta\dot{x} dt = \int_{t_1}^{t_2} m\dot{x}\delta\dot{x} dt = \int_{t_1}^{t_2} \delta \left(\frac{1}{2}m\dot{x}^2 \right) dt = \int_{t_1}^{t_2} \delta T dt.$$

We use $T = \frac{1}{2}m\dot{x}^2$ to denote the kinetic energy. Once again, the first term on the RHS cancels out because the $\delta x = 0$ at the endpoints. Integrating the second, term we get:

$$\int_{t_1}^{t_2} -\frac{dU}{dx}\delta x dt = \int_{t_1}^{t_2} -\delta U dt.$$

Refer back to the properties of the variation if you are confused on these calculations. Plugging everything back into the original equation, we get:

$$\int_{t_1}^{t_2} (\delta T - \delta U) dt = \delta \int_{t_1}^{t_2} (T - U) dt = 0.$$

Thus we define the Lagrangian $\mathcal{L} = T - U$, and

$$\delta \int_{t_1}^{t_2} \mathcal{L} dt = 0.$$

This is Hamilton's Principle.

6 Lagrange's Equations

Hamilton's Principle tells us that for the path a system takes, $x(t)$, the integral of the Lagrangian must be stationary. This problem is exactly what the Euler-Lagrange equation solves. Combining the two, we get Lagrange's Equations. Note that $\mathcal{L} = \mathcal{L}[t, x, \dot{x}]$.

Thus, the Euler-Lagrange equation tells us:

$$\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = 0.$$

7 Generalized Coordinates

Although we derived Lagrange's equation with only one coordinate, it can be applied to any coordinate x, y, z . In fact, we can use any coordinate system. In general,

$$\frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) = 0,$$

where q_i is a generalized coordinate in a system $\mathbf{r} = (q_1, q_2, \dots, q_n)$.

This also allows us to define a few more useful generalized quantities. Returning to the x coordinate, notice how

$$\frac{\partial \mathcal{L}}{\partial x} = -\frac{dU}{dx} = F_x$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{dT}{d\dot{x}} = m\dot{x} = p_x.$$

This suggests that for any generalized coordinate, we can define the following:

$$\begin{aligned} \text{ith component of generalized force} &= \frac{\partial \mathcal{L}}{\partial q_i} \\ \text{ith component of generalized momentum} &= \frac{\partial \mathcal{L}}{\partial \dot{q}_i}. \end{aligned}$$

8 Applications of Lagrange's Equations

Finally, time to solve some physics problems! (or not) Let's first look at a simple pendulum. We can describe the energies in terms of the angle θ :

$$T = \frac{1}{2}l^2\dot{\theta}^2, \text{ and } U = -mgl \cos \theta.$$

Then, the Lagrangian is

$$\mathcal{L} = T - U = \frac{1}{2}l^2\dot{\theta}^2 + mgl \cos \theta.$$

Applying Lagrange's equation, we get

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \theta} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) &= 0 \\ (-mgl \sin \theta) - \frac{d}{dt} (ml^2\dot{\theta}) &= 0 \\ -mgl \sin \theta - ml^2\ddot{\theta} &= 0 \\ g \sin \theta + l\ddot{\theta} &= 0 \end{aligned}$$

This is the same differential equation you would get by applying Newton's 2nd Law. So why use Lagrange's equation over Newton's 2nd Law? Well, it would take quite a bit of space to solve a non-trivial example, so instead I will link a handout on applying Lagrangian mechanics to a double pendulum, where the differences between Lagrangian mechanics and Newton's 2nd Law becomes much clearer.

<https://www.phys.lsu.edu/faculty/gonzalez/Teaching/Phys7221/DoublePendulum.pdf>

9 Closing Remarks

This is really only an introduction to the subject. What happens when you use coordinates that are time-dependent? What happens if the system is not conservative? These are problems that I still haven't learned about. The goal of this handout is to give some insight as to why finding equations of motion can be thought of as a minimization problem or finding stationary points in general.

10 Appendix

10.1 Partial Derivatives

It turns out that most things in the real world are described by more than one variable. For example, take a metal plate with different temperatures at different points. We have to use two coordinates to describe where a point is, whether it's rectangular coordinates or polar coordinates. We can then write a function $f(x, y)$ that gives us the temperature at the corresponding point.

One useful question to ask is about the rate of change of temperature as we walk along this metal plate. The problem is that there are infinitely many directions to walk in. Often times, it's useful to find the rates of change with respect to each of the coordinates in the coordinate system we are using, which is where partial derivatives come in.

Partial derivatives give the rates of change with respect to a single variable, and when doing calculations, we take the derivative with respect to that variable while treating all the other variables like constants.

$\frac{\partial f}{\partial x}$ is read as the partial derivative of f with respect to x .

For example, if $f(x, y) = x^2 + y$, then $\partial f / \partial x = 2x$.

10.2 And in relation to integrals

We will also use the following trick:

$$\frac{d}{dy} \int_{x_1}^{x_2} f dx = \int_{x_1}^{x_2} \frac{\partial f}{\partial y} dx.$$

The reason it turns into a partial derivative is because, before integrating, the function still depends on x , but it also must depend on y since that's what we're taking a derivative with respect to. We only want to take the derivative with respect to y so we take a partial derivative.

10.3 Multivariable Chain Rule

Recall the chain rule from single variable calculus:

$$\frac{df}{dt} = \frac{df}{dx} \frac{dx}{dt}.$$

When a function depends on multiple variables, each of those variables will have its own rate of change and we sum up the effects of the chain rule on each variable.

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

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